

11

Energy

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In this chapter we are going to focus on one of the key concepts in thermal physics, that of *energy*. What happens when energy is changed from one form to another? How much work can you get out of a quantity of heat? These are key questions to be answered. We are now beginning a study of **thermodynamics** proper, and in this chapter we will introduce the first law of thermodynamics. Before the first law, the most important concept in this chapter, we will introduce some additional ideas.

11.1 Some definitions

11.1.1 A system in thermal equilibrium

In thermodynamics, we define a **system** to be whatever part of the Universe we select for study. Near the system are its **surroundings**. We recall from Section 4.1 that a system is in *thermal equilibrium* when its macroscopic observables (such as its pressure or its temperature) have ceased to change with time. If you take a gas in a container which has been held at a certain stable temperature for a considerable period of time, the gas is likely to be in thermal equilibrium. A system in thermal equilibrium having a particular set of macroscopic observables is said to be in a particular **equilibrium state**. If however, you *suddenly* apply a lot of heat to one side of the box, then *initially at least*, the gas is likely to be in a **non-equilibrium state**.

11.1.2 Functions of state

A system is in an equilibrium state if macroscopic observable properties have fixed, definite values, independent of 'how they got there'. These properties are **functions of state** (sometimes called **variables of state**). A **function of state** is any physical quantity that has a well-defined value for each equilibrium state of the system. Thus, in thermal equilibrium these variables of state have no time dependence. Examples are volume, pressure, temperature and internal energy, and we will introduce a lot more in what follows. Examples of quantities which are not functions of state include the position of particle number 4325667, the total work done on a system and the total heat put into the system. Below, we will show in detail why work and heat are not functions of state. However, the point can be understood as follows: the

fact that your hands are warm or cold depends on their current temperature (a function of state), independently of how you got them to that temperature. For example, you can get to the same final thermodynamic state of having warm hands by different combinations of working and heating, e.g. you can end up with warm hands by rubbing them together (using the muscles in your arms to do work on them) or putting them in a toaster¹ (adding heat).

We now give a more mathematical treatment of what is meant by a function of state. Let the state of a system be described by parameters $\mathbf{x} = (x_1, x_2, \dots)$ and let $f(\mathbf{x})$ be some function of state. [Note that this could be a very trivial function, such as $f(\mathbf{x}) = x_1$, since what we've called 'parameters' are themselves functions of state. But we want to allow for more complicated functions of state which might be combinations of these 'parameters'.] Then if the system parameters change from $\mathbf{x}_i$ to $\mathbf{x}_f$, the change in f is

$$\Delta f = \int_{\mathbf{x}_i}^{\mathbf{x}_f} df = f(\mathbf{x}_f) - f(\mathbf{x}_i). \quad (11.1)$$

This only depends on the endpoints $\mathbf{x}_i$ and $\mathbf{x}_f$. The quantity df is an **exact differential** (see Appendix C.7) and functions of state have exact differentials. By contrast, a quantity which is represented by an **inexact differential** is not a function of state. The following example illustrates these kinds of differentials.

Example 11.1

Let a system be described by two parameters, x and y . Let $f = xy$ so that

$$df = d(xy) = y dx + x dy. \quad (11.2)$$

Then if (x, y) changes from $(0, 0)$ to $(1, 1)$, the change in f is given by

$$\Delta f = \int_{(0,0)}^{(1,1)} df = [xy]_{(0,0)}^{(1,1)} = (1 \times 1) - (0 \times 0) = 1. \quad (11.3)$$

This answer is independent of the exact path taken (it could be any of those shown in Fig. 11.1) because df is an exact differential.

Now consider² $dg = y dx$. The change in g when (x, y) changes from $(0, 0)$ to $(1, 1)$ along the path shown in Fig. 11.1(a) is given by

$$\Delta g = \int_{(0,0)}^{(1,1)} y dx = \int_0^1 x dx = \frac{1}{2}. \quad (11.4)$$

However when the integral is not carried out along the line $y = x$, but along the path shown in Fig. 11.1(b), it is given by

$$\Delta g = \int_{(0,0)}^{(1,0)} y dx + \int_{(1,0)}^{(1,1)} y dx = 0. \quad (11.5)$$

¹NB don't try this at home.

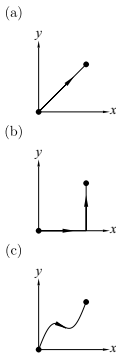


Fig. 11.1 Three possible paths between the points $(x, y) = (0, 0)$ and $(x, y) = (1, 1)$.

²We put a line through quantities such as the d in dg to signify that it is an inexact differential.

³Note that if x is taken to be volume, V , and y is taken to be pressure, p , then the quantity f is proportional to temperature, while dg is the negative of the work $dW = -p dV$. This demonstrates that temperature is a function of state and work is not.

If the integral is taken along the path shown in Fig. 11.1(c), yet another result would be obtained, but we are not going to attempt to calculate that!

Hence we find that the value of Δg depends on the path taken, and this is because dg is an inexact differential.³

Recall from Section 1.2 that functions of state can either be:

- **extensive** (proportional to system size), e.g. energy, volume, magnetization, mass, or
- **intensive** (independent of system size), e.g. temperature, pressure, magnetic field, density, energy density.

In general one can find an **equation of state** which connects functions of state: for a gas this takes the form $f(p, V, T) = 0$. An example is the equation of state for an ideal gas, $pV = nRT$, which we met in eqn 1.12.

11.2 The first law of thermodynamics

Though the idea that heat and work are both forms of energy seems obvious to a modern physicist, the idea took some getting used to. Lavoisier had, in 1789, proposed that heat was a weightless, conserved fluid called *caloric*. Caloric was a fundamental element that couldn't be created or destroyed. Lavoisier's notion 'explained' a number of phenomena, such as combustion (fuels have stored caloric which is released on burning). Rumford in 1798 realised that something was wrong with the caloric theory: heating could be produced by friction, and if you keep on drilling through a cannon barrel (to take the example that drew the problem to his attention) almost limitless supplies of heat can be extracted. Where does all this caloric come from? Mayer quantified this in 1842 with an elegant experiment in which he frictionally generated heat in paper pulp and measured the temperature rise. Joule⁴ independently performed similar experiments, but more accurately, in the period 1840–1845 (and his results became better known so that he was able to claim the credit!) Joule let a mass tied to a string slowly descend a certain height, while the other end of the string turns a paddle wheel immersed in a certain mass of water. The turning of the paddle frictionally heats the water. After a number of descents, Joule measured the temperature rise of the water. In this way he was able to deduce the 'mechanical equivalent of heat'. He also measured the heat output of a resistor (which, in modern units, is equal to $I^2 R$, where I is the current and R the resistance). He was able to show that the same heat was produced for the same energy used, independent of the method of delivery. This implied that heat is a form of energy. Joule's experiments therefore consigned the caloric theory of heat to a footnote in history.

However, it was Mayer and later Helmholtz who elevated the experimental observations into a grand principle, which we can state as follows:

⁴The S.I. unit of energy is named after Joule. $1\text{ J} = 1\text{ N m}$. Older units are still in use in some places: 1 calorie is defined as the energy required to raise 1 g of water by 1°C (actually from 14.5°C to 15.5°C at sea level) and $1\text{ cal} = 4.184\text{ J}$. The energy contained in food is usually measured in kilocalories (kcal), where $1\text{ kcal} = 1000\text{ cal}$. Older books sometimes used the erg: $1\text{ erg} = 10^{-7}\text{ J}$. The British thermal unit (Btu) is an archaic unit, no longer commonly used in Britain: $1\text{ Btu} = 1055\text{ J}$. The foot-pound is $1\text{ ft lb} = 1.356\text{ J}$. Electricity bills often record energy in kilowatt hours ($1\text{ kWh} = 3.6\text{ MJ}$). Useful in atomic physics is the electron Volt: $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$.

The first law of thermodynamics:

Energy is conserved and heat and work are both forms of energy.

A system has an **internal energy** U which is the sum of the energy of all the internal degrees of freedom that the system possesses. U is a function of state because it has a well-defined value for each equilibrium state of the system. We can change the internal energy of the system by heating it or by doing work on it. The heat Q and work W are not functions of state since they concern the manner in which energy is delivered to (or extracted from) the system. After the event of delivering energy to the system, you have no way of telling which of Q or W was added to (or subtracted from) the system by examining the system's state.

The following analogy may be helpful: your personal bank balance behaves something like the internal energy U in that it acts like a function of state of your finances; cheques and cash are like heat and work in that they both result in a change in your bank balance, but after they have been paid in, you can't tell by simply looking at the value of your bank balance by which method the money was paid in.

The change in internal energy U of a system can be written

$$\Delta U = \Delta Q + \Delta W, \quad (11.6) \quad (b)$$

where ΔQ is the heat supplied to the system and ΔW is the work done on the system. Note the convention: ΔQ is positive for heat supplied to the system; if ΔQ is negative, heat is extracted from the system; ΔW is positive for work done on the system; if ΔW is negative, the system does work on its surroundings.

We define a **thermally isolated system** as a system which cannot exchange heat with its surroundings. In this case we find that $\Delta U = \Delta W$, because no heat can pass in or out of a thermally isolated system.

For a differential change, we write eqn 11.6 as

$$dU = dQ + dW, \quad (11.7)$$

where dW and dQ are inexact differentials.

The work done on stretching a wire by a distance dx with a tension F is (see Fig. 11.2(a))

$$dW = F dx. \quad (11.8)$$

The work done by compressing a gas (pressure p , volume V) by a piston can be calculated in a similar fashion (see Fig. 11.2(b)). In this case the force is $F = pA$, where A is the area of the piston, and $A dx = -dV$, so that

$$dW = -p dV. \quad (11.9)$$

In this equation, the negative sign ensures that the work dW done on the system is positive when dV is negative, i.e. when the gas is being compressed.

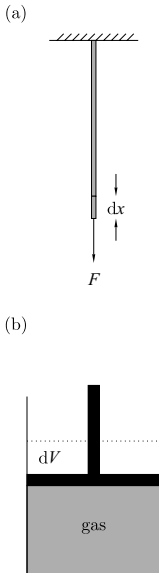


Fig. 11.2 (a) The work done stretching a wire by a distance dx is $F dx$. (b) The work done compressing a gas is $-p dV$.

It turns out that eqn 11.9 is only strictly true for a reversible change, a point we will explain further in Section 12.1. The idea is that if the piston is not frictionless, or if you move the piston too suddenly and generate shock waves, you will need to do more work to compress the gas because more heat is dissipated in the process.

11.3 Heat capacity

We now want to understand in greater detail how adding heat can change the internal energy of gas. In general, the internal energy will be a function of temperature and volume, so that we can write $U = U(T, V)$. Hence a small change in U can be related to changes in T and V by

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV. \quad (11.10)$$

Rearranging eqn 11.7 with eqn 11.9 yields

$$dQ = dU + p dV, \quad (11.11)$$

and now using eqn 11.10 we have that

$$dQ = \left(\frac{\partial U}{\partial T}\right)_V dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] dV. \quad (11.12)$$

We can divide eqn 11.12 by dT to obtain

$$\frac{dQ}{dT} = \left(\frac{\partial U}{\partial T}\right)_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] \frac{dV}{dT}, \quad (11.13)$$

which is valid for any change in T or V . However, what we want to know is what is the amount of heat we have to add to effect a change of temperature under certain constraints. The first constraint is that of keeping the volume constant. We recall the definition of the heat capacity at constant volume C_V (see Section 2.2, eqn 2.6) as

$$C_V = \left(\frac{\partial Q}{\partial T}\right)_V. \quad (11.14)$$

From eqn 11.13, this constraint knocks out the second term and implies that

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V. \quad (11.15)$$

The heat capacity at constant pressure is then, using eqns 2.7 and 11.13, given by

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_p \quad (11.16)$$

$$= \left(\frac{\partial U}{\partial T}\right)_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] \left(\frac{\partial V}{\partial T}\right)_p \quad (11.17)$$

so that

$$C_p - C_V = \left[\left(\frac{\partial U}{\partial V} \right)_T + p \right] \left(\frac{\partial V}{\partial T} \right)_p. \quad (11.18)$$

Recall from Section 2.2 that heat capacities are measured in JK^{-1} and refer to the heat capacity of a certain quantity of gas. We will sometimes wish to talk about the heat capacity per mole of gas, or sometimes the heat capacity per mass of gas. We will use small c for the latter, known as the *specific heat capacities*:

$$c_V = \frac{C_V}{M} \quad (11.19)$$

$$c_p = \frac{C_p}{M} \quad (11.20)$$

$$(11.21)$$

where M is the mass of the material. Specific heat capacities are measured in $\text{JK}^{-1} \text{kg}^{-1}$.

Example 11.2

Heat capacity of an ideal monatomic gas

For an ideal monatomic gas, the internal energy U is due to the kinetic energy, and hence $U = \frac{3}{2}RT$ per mole (see eqn 5.17; this result arises from the kinetic theory of gases). This means that U is only a function of temperature. Hence

$$\left(\frac{\partial U}{\partial V} \right)_T = 0. \quad (11.22)$$

The equation of state for 1 mole of ideal gas is

$$pV = RT, \quad (11.23)$$

so that

$$V = \frac{RT}{p}, \quad (11.24)$$

and hence

$$\left(\frac{\partial V}{\partial T} \right)_p = \frac{R}{p}, \quad (11.25)$$

and hence using eqns 11.18, 11.22 and 11.25 we have that

$$C_p - C_V = \left[\left(\frac{\partial U}{\partial V} \right)_T + p \right] \left(\frac{\partial V}{\partial T} \right)_p = R. \quad (11.26)$$

Because $U = \frac{3}{2}RT$, we therefore have that

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{3}{2}R \text{ per mole}, \quad (11.27)$$

and

$$C_p = C_V + R = \frac{5}{2}R \text{ per mole}. \quad (11.28)$$

Example 11.3

Is it always true that $dU = C_V dT$?

Solution: No, in general eqn 11.10 and eqn 11.15 imply that

$$dU = C_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV. \quad (11.29)$$

For an ideal gas, $\left(\frac{\partial U}{\partial V} \right)_T = 0$ (eqn 11.22) so it is true that

$$dU = C_V dT, \quad (11.30)$$

but for non-ideal gases, $\left(\frac{\partial U}{\partial V} \right)_T \neq 0$ and hence $dU \neq C_V dT$.

The ratio of C_p to C_V turns out to be a very useful quantity (we will see why in the following chapter) and therefore we give it a special name. Hence, we define the **adiabatic index**⁵ γ as the ratio of C_p and C_V , so that

$$\gamma = \frac{C_p}{C_V}. \quad (11.31)$$

The reason for the name will become clear in the following chapter.

Example 11.4

What is γ for an ideal monatomic gas?

Solution: Using the results from the previous example⁶

$$\gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V} = \frac{5}{3}. \quad (11.32)$$

⁵ γ is sometimes called the **adiabatic exponent**.

⁶If the gas is not monatomic, γ can take a different value; see Section 19.2.

Example 11.5

Assuming $U = C_V T$ for an ideal gas, find (i) the internal energy per unit mass and (ii) the internal energy per unit volume.

Solution: Using the ideal gas equation $pV = Nk_B T$ and the density $\rho = Nm/V$ (where m is the mass of one molecule), we find that

$$\frac{p}{\rho} = \frac{k_B T}{m}. \quad (11.33)$$

Using eqn 11.32, we have that the heat capacity per mole is given by

$$C_V = \frac{R}{\gamma - 1}. \quad (11.34)$$

Hence, we can write that the internal energy for one mole of gas is

$$U = C_V T = \frac{RT}{\gamma - 1} = \frac{N_A k_B T}{\gamma - 1}. \quad (11.35)$$

The molar mass is mN_A , and so dividing eqn 11.35 by the molar mass, yields $\tilde{u}$, the internal energy per unit mass, given by

$$\tilde{u} = \frac{p}{\rho(\gamma - 1)}. \quad (11.36)$$

Multiplying $\tilde{u}$ by the density ρ gives u , the internal energy per unit volume, as

$$u = \rho \tilde{u} = \frac{p}{\gamma - 1}. \quad (11.37)$$

Chapter summary

- Functions of state have exact differentials.
- The first law of thermodynamics states that ‘energy is conserved and heat is a form of energy’.
- $dU = dW + dQ$.
- For a reversible change, $dW = -p dV$.
- $C_V = \left(\frac{\partial Q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$.
- $C_V = \left(\frac{\partial Q}{\partial T}\right)_P$ and $C_p - C_V = R$ for a mole of ideal gas.
- The adiabatic index is $\gamma = C_p/C_V$.

Exercises

(11.1) One mole of ideal monatomic gas is confined in a cylinder by a piston and is maintained at a constant temperature T_0 by thermal contact with a heat reservoir. The gas slowly expands from V_1 to V_2 while being held at the same temperature T_0 . Why does the internal energy of the gas not change? Calculate the work done by the gas and the heat flow into the gas.

(11.2) Show that, for an ideal gas,

$$\frac{R}{C_V} = \gamma - 1 \quad (11.38)$$

and

$$\frac{R}{C_p} = \frac{\gamma - 1}{\gamma}, \quad (11.39)$$

where C_V and C_p are the heat capacities per mole.

(11.3) Consider the differential

$$dz = 2xy dx + (x^2 + 2y) dy. \quad (11.40)$$

Evaluate the integral $\int_{(x_1, y_1)}^{(x_2, y_2)} dz$ along the paths consisting of straight-line segments

(i) $(x_1, y_1) \rightarrow (x_2, y_1)$ and then $(x_2, y_1) \rightarrow (x_2, y_2)$.

(ii) $(x_1, y_1) \rightarrow (x_1, y_2)$ and then $(x_1, y_2) \rightarrow (x_2, y_2)$.

Is dz an exact differential?

- (11.4) In polar coordinates, $x = r \cos \theta$ and $y = r \sin \theta$. The definition of x implies that

$$\frac{\partial x}{\partial r} = \cos \theta = \frac{x}{r}. \quad (11.41) \quad (11.5)$$

But we also have $x^2 + y^2 = r^2$, so differentiating with respect to r gives

$$2x \frac{\partial x}{\partial r} = 2r \quad \implies \quad \frac{\partial x}{\partial r} = \frac{r}{x}. \quad (11.42)$$

But eqns 11.41 and 11.42 imply that

$$\frac{\partial x}{\partial r} = \frac{\partial r}{\partial x}. \quad (11.43)$$

What's gone wrong?

(11.5) In the comic song by Flanders and Swann about the laws of thermodynamics, they summarize the first law by the statement:

Heat is work and work is heat

Is that a good summary?